ARC-SPRUCE: Post-Quantum Blind Issuance and Anonymous Rate-Limited Credentials from Lattices

Jonas Lazard
jonas.lazard@student-cs.fr
DTU Compute
Kongens Lyngby, Denmark

Carsten Baum
cabau@dtu.dk
DTU Compute
Kongens Lyngby, Denmark

ABSTRACT

Blind issuance protocols and anonymous credentials are useful for privacy-preserving online authentication systems. ARC-SPRUCE gives a post-quantum issuer-view-hiding blind-issuance construction and a conditional anonymous rate-limiting credential construction from lattice assumptions and hash-based zero knowledge. The construction uses an IntGenISIS-style committed-message BB-tran issuance layer: the holder sends a target-shaped Ajtai/BDLOP-style commitment to the semantic message, and the issuer signs the sum of an issuer-sampled rational-hash target and that commitment. Message privacy is computational, from Module-LWE hiding of the commitment and zero knowledge of the issuance proof; binding is from Module-SIS. The remaining layers are the same Gaussian preimage-sampling interface, SmallWood NIZKs, and Poseidon2-style PRF assumption.

ACM Reference Format:
Jonas Lazard and Carsten Baum. 2026. ARC-SPRUCE: Post-Quantum Blind Issuance and Anonymous Rate-Limited Credentials from Lattices. In *Proceedings of ACM Conference (Conference'17).* ACM, New York, NY, USA, 15 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

Anonymous credentials (AC) [2–4] allow parties to convey certified information about themselves in a privacy-preserving way. The usual roles are issuers, holders, and verifiers. A holder obtains attributes certified by an issuer and later proves statements about those attributes to a verifier without revealing the credential itself. The central security goals are unforgeability and unlinkability: verifiers should only accept credentials derived from an issuer, while issuers and verifiers should not be able to link a presentation to an issuance session or to another presentation.

Certain AC use cases, such as anonymous rate limiting [10, 11], require that a credential be usable only a bounded number of times in a given context, for example per epoch or per service. Existing classical constructions can provide compact credentials and presentations, but practically efficient post-quantum ACs remain much less developed. This gap is relevant for digital identity systems, including wallet deployments that will eventually require post-quantum migration.

Building ACs. Anonymous credentials are often built from a proof-friendly signature and a compatible zero-knowledge proof. The holder obtains a signature on hidden attributes and later proves, in zero knowledge, knowledge of a valid message-signature pair. A generic post-quantum combination of standard signatures, commitments, and zero-knowledge proofs is currently too large for the settings targeted here. We therefore combine a proof-friendly lattice signature, a compact lattice commitment term, and SmallWood zero-knowledge proofs in a relation designed to remain low degree.

1.1 Contributions

We construct a post-quantum issuer-view-hiding issuance design and a conditional anonymous rate-limited credential design from lattice assumptions and hash-based zero knowledge. The construction uses proof-friendly lattice signatures from vanishing SIS with rational hashing [5], an NTRU-style Gaussian preimage-sampling candidate based on Antrag [6], and the SmallWood proof system for relatively small statements [7]. The DKLW import is used for the non-blind hash-and-preimage signature layer. The interactive issuance protocol is treated separately as a correct and issuer-view-hiding committed-message issuance layer, with one-more unforgeability left as an explicit assumption in the ARC soundness decomposition. Rate limiting is implemented by signing a hidden PRF key and proving correct tag derivation during presentation.

The issuance layer follows the IntGenISIS/BLNS/DKLW shape. The holder's semantic signed message is

$$M := \mathbf{m} \| \mathbf{k},$$

where $\mathbf{m}$ is the attribute vector and $\mathbf{k}$ is the secret PRF key. The holder sends a target-shaped Ajtai/BDLOP-style commitment

$$c = C_M M + A_s s + e \in R_q^{n_c}.$$

The issuer then samples the BB-tran rational-hash characteristic and randomness independently of M. With $\mu_{\text{sig}} \in R_q^{\ell_{\mu_{\text{sig}}}}$, $x_0 \in R_q^{\ell_{x_0}}$, and $x_1 \in R_q$, the rational-hash part is

$$h_{\mu_{\text{sig}},\chi_{\text{sig}}}(B) = B_0 + B_1\mu_{\text{sig}} + B_2 x_0 + Z, \qquad (B_3 - \mathbf{1}_n x_1) \odot Z = 1_n,$$

where $\chi_{\text{sig}} = (x_0, x_1)$. The signed target is

$$T = h_{\mu_{\text{sig}},\chi_{\text{sig}}}(B) + c.$$

The preimage signature satisfies $\mathbf{Au} = T$. Thus the semantic message appears only in the computationally hiding commitment term, not directly in the BB-tran rational-hash characteristic.

The issuer-view privacy statement is computational. It relies on Module-LWE hiding of the commitment term and zero knowledge

Conference'17, July 2017, Washington, DC, USA

ACM ISBN 978-x-xxxx-xxxx-x/YY/MM…$15.00
https://doi.org/10.1145/nnnnnnn.nnnnnnn

of the pre-sign proof; binding of committed openings is reduced to Module-SIS for the block matrix $[C_M \mid A_s \mid I_{n_c}]$. This is a local target-shaped specialization of the Ajtai/BDLOP/ABDLOP commitment precedent used in lattice zero-knowledge protocols [1, 9]; we do not claim that this exact compact form is copied verbatim from those sources.

1.2 Technical overview

The semantic signed message is written throughout as

$$M = \mathbf{m} \| \mathbf{k},$$

with an explicit encoding assumption $\ell_M = 1$ for the first target parameter set. The DKLW rational-hash message or characteristic is written μ_{sig} to avoid confusion with the semantic message.

Issuance. The holder samples $\mathbf{k} \xleftarrow{\$} \mathcal{K}_{\mathsf{key}}$, forms $M = \mathbf{m} \| \mathbf{k}$, samples commitment randomness $s \leftarrow \chi_s^{k_s}$ and error $e \leftarrow \chi_e^{n_c}$, and sends

$$c = C_M M + A_s s + e$$

together with a pre-sign NIZK π_{pre}. This proof shows knowledge of a bounded opening (M, s, e), correct encoding of $M = \mathbf{m} \| \mathbf{k}$, key-space membership for $\mathbf{k}$, and any issuance policy constraints on $\mathbf{m}$. It does not prove the BB-tran inverse relation or the final target computation.

The issuer verifies π_{pre}, samples

$$\mu_{\mathsf{sig}} \in R_q^{\ell_{\mu_{\mathsf{sig}}}}, \qquad x_0 \in R_q^{\ell_{x_0}}, \qquad x_1 \in R_q,$$

rejects and resamples if $B_3 - \mathbf{1}_n x_1$ is not invertible, sets

$$Z = (B_3 - \mathbf{1}_n x_1)^{-1}, \qquad T = B_0 + B_1 \mu_{\mathsf{sig}} + B_2 x_0 + Z + c,$$

and samples $\mathbf{u} \leftarrow \mathsf{SampPre}(\mathsf{td}, T)$ such that $\mathbf{A}\mathbf{u} = T$. The issuer returns $(\mathbf{u}, \mu_{\mathsf{sig}}, x_0, x_1)$. The holder stores the credential witness

$$(\mathbf{u}, M, \mathbf{m}, \mathbf{k}, s, e, \mu_{\mathsf{sig}}, x_0, x_1).$$

If M already carries $\mathbf{m}$ and $\mathbf{k}$ in the implementation, $\mathbf{m}$ can be omitted from storage.

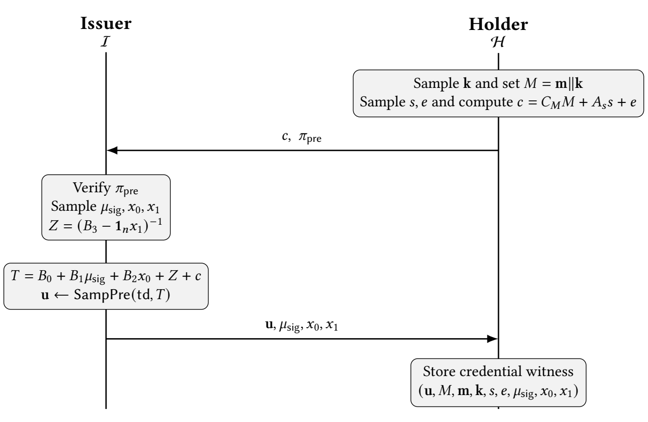

Figure 1: Issuance flow for the committed-message IntGenISIS-style construction.

Showing. To present a credential, the holder chooses a nonce, computes $\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce})$, reconstructs $Z = (B_3 - \mathbf{1}_n x_1)^{-1}$, and proves knowledge of $\mathbf{u}, M, \mathbf{m}, \mathbf{k}, s, e, \mu_{\mathsf{sig}}, x_0, x_1, Z$ such that

$$\begin{aligned}
&(B_3 - \mathbf{1}_n x_1) \odot Z = 1_n,\\
&\mathbf{A}\mathbf{u} = B_0 + B_1 \mu_{\mathsf{sig}} + B_2 x_0 + Z + C_M M + A_s s + e,\\
&M = \mathbf{m} \| \mathbf{k}, \qquad \mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}).
\end{aligned}$$

The verifier sees only $(\mathsf{nonce}, \mathsf{tag}, \pi_{\mathsf{Show}})$ and the public parameters. The issuance commitment c is not revealed during showing.

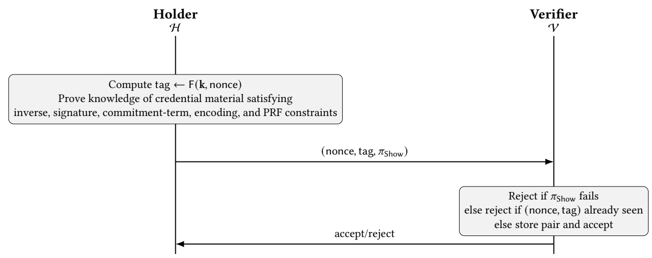

Figure 2: Showing flow. The proof hides all credential material, including the commitment opening and rational-hash randomness.

SmallWood proofs. We use SmallWood [7] as the NIZK for the issuance and showing relations. The new relations remain proof-friendly: the commitment equation and the signature equation are linear over the ring, while the rational-hash inverse equation is bilinear. The PRF checks retain their existing Poseidon2-style non-linear constraints. Section 5 states the semantic PACS relations and the coefficient-domain consequences used by the security arguments; Appendix C records the schematic row inventories and proof-stack details.

2 PRELIMINARIES

This section fixes notation and recalls the formal interfaces used by the construction: the lattice commitment term, the BB-tran rational hash, non-interactive zero-knowledge arguments, the issuance and ARC syntax, and the PRF. We also record the imported non-blind signature-security statements from Dubois–Klooß–Lai–Woo (DKLW) that are used later for the hash-and-preimage signature layer.

2.1 Notation

Let λ denote the security parameter and let $\mathsf{negl}(\lambda)$ be any positive function negligible in λ. We write PPT for probabilistic polynomial time. Vectors are written in bold lower-case letters and matrices in bold upper-case letters.

We work over the cyclotomic ring

$$R = \mathbb{Z}[X]/(X^N + 1)$$

for power-of-two N, and its quotient $R_q = R/qR$. We identify R_q with polynomials of degree $< N$ with coefficients in $\mathbb{Z}_q$. Whenever a field element is interpreted as a signed integer, we use its signed representative in $[-q/2, q/2]$. For $x \in R_q$, the notation $\|x\|_\infty \leq B$

means that every coefficient of x lies in $[-B, B]$; the same convention is used component-wise for vectors over R_q.

The semantic signed message is

$$M := \mathbf{m} \| \mathbf{k} \in R_q^{\ell_M},$$

where $\mathbf{m}$ encodes holder attributes and $\mathbf{k}$ is the secret PRF key used for rate limiting. The first target parameter set uses $\ell_M = 1$, so the encoding of $\mathbf{m}\|\mathbf{k}$ into one ring polynomial is an explicit parameter assumption. If a deployment needs a longer semantic message, ℓ_M must be increased and all commitment and SmallWood row counts recomputed.

The PRF key is sampled from a finite key space $\mathcal{K}_{\mathsf{key}} \subseteq \mathbb{F}_q^{\ell_{\mathsf{key}}}$ fixed by the public parameters. In the bounded-key instantiation, we require

$$\mathcal{K}_{\mathsf{key}} \subseteq \{\mathbf{k} \in \mathbb{F}_q^{\ell_{\mathsf{key}}} : \|\mathbf{k}\|_\infty \le \beta_{\mathsf{msg}}\},$$

and the PRF assumption is made for a uniform key from this same key space. The public coefficient bound for holder attributes and commitment randomness is denoted β_{msg}, and the short-preimage bound for signatures is denoted $\beta_{\mathtt{sig}}$.

Whenever the CRT decomposition $R_q \cong \prod_i F_i$ is used, we write

$$(x_i)_i \odot (y_i)_i := (x_i y_i)_i, \qquad (x_i)_i \oslash (y_i)_i := (x_i y_i^{-1})_i$$

whenever every $y_i \neq 0$. Thus $\odot$ and $\oslash$ denote slotwise multiplication and slotwise division in CRT coordinates.

2.2 Target-shaped lattice commitment

The issuance protocol uses a compact target-shaped lattice commitment to the semantic message. The shape is a local specialization of the Ajtai/BDLOP/ABDLOP commitment lineage used in lattice zero-knowledge protocols [1, 9]. The source precedent is a lattice commitment with hiding based on Module-LWE-type pseudorandomness and binding based on Module-SIS-type short-relation hardness. The exact compact one-target form below is the ARC-SPRUCE specialization.

Let

$$M \in R_q^{\ell_M}, \qquad s \in R^{k_s}, \qquad e \in R^{n_c}.$$

The public matrices are

$$C_M \in R_q^{n_c \times \ell_M}, \qquad A_s \in R_q^{n_c \times k_s},$$

and the commitment is

$$c = C_M M + A_s s + e \in R_q^{n_c}. \tag{1}$$

Equivalently, with

$$C = [C_M \mid A_s \mid I_{n_c}],$$

this is the block equation

$$c = C\,[M; s; e].$$

The identity block is intentional: in the fully split ring, a uniformly random single ring element is not invertible with overwhelming probability for the parameter scales considered here, so no random invertible error block is used.

Definition 2.1 (Opening relation for the commitment). $\mathsf{Open}(c; M, s, e) = 1$ if and only if (1) holds in $R_q^{n_c}$, M is a valid encoding of the semantic message, the key-space constraints inside M hold, and the coefficients of opened values satisfy the prescribed bounds. For the target set used in Section 6, $\ell_M = 1$, $k_s = 2$, $n_c = 1$, and the coefficients of s, e are bounded by $\beta_{\mathsf{msg}} = 8$.

Definition 2.2 (Module-LWE hiding assumption). The assumption $\mathsf{MLWE\text{-}Hiding}_{R,q,n_c,k_s,\chi_s,\chi_e}$ says that for $A_s \xleftarrow{\$} R_q^{n_c \times k_s}$, $s \leftarrow \chi_s^{k_s}$, $e \leftarrow \chi_e^{n_c}$, and $U \xleftarrow{\$} R_q^{n_c}$,

$$(A_s, A_s s + e) \approx_c (A_s, U).$$

Lemma 2.3 (Computational hiding of the target-shaped commitment). *Under Assumption 2.2, commitments to any two fixed semantic messages M_0, M_1 are computationally indistinguishable.*

Proof. For fixed M, the distribution of $c = C_M M + A_s s + e$ is a deterministic offset $C_M M$ plus $A_s s + e$. By Assumption 2.2, this is computationally indistinguishable from the same offset plus a uniform U. Since $U + C_M M$ is uniform over $R_q^{n_c}$, the resulting distribution is independent of M. Applying the argument to M_0 and M_1 gives the claim. □

Definition 2.4 (Module-SIS binding assumption). Let

$$L_{\mathsf{bind}} = \ell_M + k_s + n_c.$$

The assumption $\mathsf{MSIS\text{-}Bind}_{R,q,n_c,L_{\mathsf{bind}},\beta_{\mathsf{bind}}}$ says that for uniform C_M, A_s, no PPT adversary can find a non-zero vector $y \in R^{L_{\mathsf{bind}}}$ with $\|y\|_\infty \le \beta_{\mathsf{bind}}$ such that

$$[C_M \mid A_s \mid I_{n_c}]y = 0 \pmod q,$$

except with negligible probability.

Lemma 2.5 (Binding of the target-shaped commitment). *If the same commitment c has two distinct bounded openings (M, s, e) and (M', s', e'), then their difference gives a short non-zero solution to the Module-SIS instance of Assumption 2.4. If all opened coefficients are bounded by β_{msg}, then the difference has coefficient bound $\beta_{\mathsf{bind}} = 2\beta_{\mathsf{msg}}$ for those opened coordinates.*

Proof. Subtracting the two equations $c = C_M M + A_s s + e = C_M M' + A_s s' + e'$ yields

$$C_M(M - M') + A_s(s - s') + (e - e') = 0.$$

Since the openings are distinct, the block vector $(M-M', s-s', e-e')$ is non-zero. Its coefficient bound is at most twice the opening bound, and it is a Module-SIS solution for $[C_M \mid A_s \mid I_{n_c}]$. □

2.3 Lattice trapdoors and the committed-message BB-tran target

We recall the interface used by the hash-and-preimage signature layer. Additional lattice-assumption syntax, such as vSIS, hinted-vSIS, $\mathrm{GenISIS}_f$, and $\mathrm{IntGenISIS}_f$, is deferred to the appendix.

Definition 2.6 (Admissible lattice trapdoor suite). A lattice trapdoor suite over R_q consists of PPT algorithms

$$(\mathsf{Setup}, \mathsf{TrapGen}, \mathsf{SampPre})$$

together with a public bound $\beta_{\mathtt{sig}}$. We say that the suite is admissible for the parameters used in this paper if the following interface holds.

- $\mathsf{pp_{td}} \leftarrow \mathsf{Setup}(1^\lambda, \mathsf{pp})$ outputs dimensions (n, m), modulus q, sampler parameters, and the accepted shortness bound $\beta_{\mathtt{sig}}$.

- $(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(\mathsf{pp}_{\mathsf{td}})$ outputs a public matrix $\mathbf{A} \in R_q^{n\times m}$ together with a trapdoor td, and the distribution of $\mathbf{A}$ is statistically or computationally close to uniform over $R_q^{n\times m}$ as required by the invoked signature theorem.
- On input $(\mathsf{pp}_{\mathsf{td}}, \mathsf{td}, T)$ with $T \in R_q^n$, the sampler $\mathsf{SampPre}(\mathsf{td}, T)$ outputs $\mathbf{u} \in R^m$ such that $\mathbf{A}\mathbf{u} = T$ in R_q^n, except with negligible failure probability, and $\|\mathbf{u}\|_\infty \leq \beta_{\mathtt{sig}}$ except with negligible probability.
- The distributional property required by the DKLW hash-and-preimage signature theorem holds for the produced preimages at the chosen Gaussian width and public bound.

This definition is an interface assumption for the signature-layer import. Appendix B describes the NTRU/Antrag-based candidate used for parameter tuning, but Antrag alone is not cited here as a proof of this full admissible-suite interface.

BB-tran translation. DKLW's BB-tran row has message u, randomness (x_0, x_1), and rational function

$$(u^\top, x_0^\top)\tilde{b}_0 + \frac{1}{\tilde{b}_1 - x_1}$$

(Table 1 of DKLW). In the committed-message construction, the rational-hash characteristic is issuer sampled and is not the semantic message. We use

$$u \leftrightarrow \mu_{\text{sig}}, \qquad x_0 \leftrightarrow x_0, \qquad x_1 \leftrightarrow x_1.$$

The public description is $B = (B_0, B_1, B_2, B_3)$, with $B_0, B_3 \in R_q^n$ and public linear maps B_1, B_2 into R_q^n. With

$$\chi_{\text{sig}} = (x_0, x_1),$$

define

$$h_{\mu_{\text{sig}},\chi_{\text{sig}}}(B) := B_0 + B_1\mu_{\text{sig}} + B_2x_0 + Z, \qquad Z = (B_3 - \mathbf{1}_n x_1)^{-1}. \quad (2)$$

The input is admissible if $B_3 - \mathbf{1}_n x_1 \in R_q^{\times n}$, equivalently if every CRT slot of every coordinate is non-zero. The witness form is

$$(B_3 - \mathbf{1}_n x_1) \odot Z = 1_n, \qquad T = B_0 + B_1\mu_{\text{sig}} + B_2x_0 + Z + C_M M + A_s s + e. \quad (3)$$

The semantic message M appears only through the commitment term.

Lemma 2.7 (Eliminated form under admissibility). *Let $B_3 - \mathbf{1}_n x_1 = (d_i)_i$ and $T - B_0 - B_1\mu_{\text{sig}} - B_2x_0 - C_M M - A_s s - e = (z_i)_i$ in CRT coordinates. If $d_i \neq 0$ for all CRT slots, then*

$$T = h_{\mu_{\text{sig}},\chi_{\text{sig}}}(B) + C_M M + A_s s + e$$

if and only if

$$(B_3 - \mathbf{1}_n x_1) \odot \big(T - B_0 - B_1\mu_{\text{sig}} - B_2x_0 - C_M M - A_s s - e\big) = 1_n.$$

Proof. In each CRT slot, the statement is $z_i = d_i^{-1}$, which is equivalent to $d_i z_i = 1$ when $d_i \neq 0$. Reassembling the slots proves the equivalence. □

2.4 Signatures and imported non-blind security

The issuance layer builds on the non-blind vSIS hash-and-preimage signature layer from DKLW. We record the standard syntax and the conditional import used by this paper.

Definition 2.8 (Signature scheme). A signature scheme for a message space $\mathcal{M}$ is a tuple of PPT algorithms

$$\Sigma = (\mathsf{Setup}, \mathsf{KG}, \mathsf{Sign}, \mathsf{Verify}).$$

It is correct if honestly generated signatures verify except with negligible probability.

Definition 2.9 (Strong existential unforgeability). Let Σ be a signature scheme. We use the standard notions of strong existential unforgeability under chosen-message attack (sEUF-CMA), selective-message attack (sEUF-SMA), and strong selective unforgeability under selective-message attack (sSUF-SMA), exactly as in the DKLW formalization.

Definition 2.10 (vSIS-based hash-and-preimage signature). Let TDRH be a trapdoor rational hash scheme with parameters $(B, \mathcal{M}, \mathcal{X}, \delta, \beta_{\mathtt{sig}})$. The associated vSIS signature scheme signs a DKLW characteristic μ by sampling randomness χ, computing $T = \mathsf{Eval}(\mathsf{pp}_{\mathsf{td}}, \mu, \chi)$, and sampling $\mathbf{u} \leftarrow \mathsf{SampPre}(\mathsf{td}, T)$. Verification checks $\chi \in \mathcal{X}$, admissibility, $\|\mathbf{u}\|_\infty \leq \beta_{\mathtt{sig}}$, and $\mathbf{A}\mathbf{u} = T$.

Theorem 2.11 (Specialized DKLW security import). *Let Σ_{tran} be the non-blind vSIS signature scheme obtained from Definition 2.10 by instantiating the rational family with the BB-tran form. Assume the admissible trapdoor-suite interface of Definition 2.6, the DKLW predicate and denominator-bound conditions, the required strong-linear-independence condition for the instantiated rational functions, the required norm-compatibility conversion between DKLW's norm convention and the public bound $\beta_{\mathtt{sig}}$, negligible inadmissibility errors, and the corresponding strong hinted-vSIS hypothesis. Then Σ_{tran} is strongly existentially unforgeable under selective-message attack.*

Proof. This is the BB-tran specialization of the DKLW selective-query security theorem for their generic vSIS-based signature family. The statement here is conditional on the DKLW side conditions being satisfied by the concrete ARC-SPRUCE parameters. □

Corollary 2.12 (Conditional imported route to sEUF-CMA). *Suppose, in addition, that the exact hypotheses of the DKLW $\mathsf{GenISIS}_f/\mathsf{IntGenISIS}_f$ lifting theorem for BB-tran are met, including the required power-of-two cyclotomic setting, the modulus and message-bound condition $q/2 > \gamma$, the dimension condition $m = n\log q + \omega(\log\lambda)$, the smoothing lower bound on the Gaussian parameter, and the theorem's stated bound loss. Then the same non-blind signature layer obtains the corresponding strong existential unforgeability guarantee under adaptive chosen-message attack.*

Proof. This is a conditional import of DKLW's GenISIS/IntGenISIS lifting theorem. It concerns only the non-blind hash-and-preimage signature layer and does not imply one-more unforgeability of the interactive issuance protocol. □

Remark 2.13 (Scope of the import). Theorems 2.11 and 2.12 concern the non-blind hash-and-preimage signature layer only. DKLW's IntGenISIS experiment natively includes a linear committed-message term; ARC-SPRUCE uses that shape for issuance, but this paper does not claim a completed proof of one-more unforgeability for the interactive issuance protocol.

2.5 Non-interactive zero-knowledge arguments of knowledge

We use the NIZK-ARK syntax of the SmallWood framework instantiated in the random-oracle model.

Definition 2.14 (NIZKs in the random-oracle model)**.** Let $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ be a family of relations. A transparent non-interactive zero-knowledge argument of knowledge for $\mathcal{R}$ in the random-oracle model, with oracle RO, is a pair of algorithms $(\mathsf{Prove}^{\mathsf{RO}}, \mathsf{Verify}^{\mathsf{RO}})$ satisfying completeness, straightline-extractable knowledge soundness, and zero knowledge. We use the notation

$$\pi \leftarrow \mathsf{Prove}^{\mathsf{RO}}(w \mid \mathcal{R}, x), \qquad \mathsf{Verify}^{\mathsf{RO}}(\pi \mid \mathcal{R}, x) \in \{0, 1\}.$$

We instantiate these proofs through the SmallWood PACS/PIOP framework compiled with Fiat–Shamir in the random-oracle model.

2.6 Issuance syntax

We keep the blind-signature-style syntax, but the protocol is stated as a committed-message issuance layer.

Definition 2.15 (Issuance syntax)**.** The issuance layer is a tuple

$$\mathsf{BSig} = (\mathsf{Setup}, \mathsf{IssuerKeyGen}, \mathsf{Issue}, \mathsf{Verify})$$

where Setup generates public parameters, IssuerKeyGen generates issuer keys $(\mathsf{vk}, \mathsf{sk})$, Issue is an interactive protocol between issuer and holder, and $\mathsf{Verify}(\mathsf{vk}, M, \mathsf{sig})$ checks raw extracted witnesses.

Definition 2.16 (Issuance correctness)**.** The issuance layer is correct if honest execution produces a stored witness that satisfies the verification relation except with negligible probability.

Definition 2.17 (Issuer-view hiding)**.** The issuance layer is issuer-view hiding if the honest issuer's view in two executions with different hidden semantic messages is computationally indistinguishable, subject to the stated commitment-hiding and zero-knowledge assumptions.

Definition 2.18 (One-more unforgeability)**.** One-more unforgeability for the interactive committed-message issuance layer says that any PPT adversary participating in at most Q issuance interactions with the honest issuer cannot output $Q + 1$ valid extracted credential witnesses on distinct semantic messages, except with negligible probability.

Remark 2.19 (Scope of the issuance layer)**.** The protocol studied in Section 3 is used as an issuer-view-hiding committed-message issuance protocol together with a raw post-issuance verification relation. One-more unforgeability remains a separate assumption in the ARC soundness decomposition.

2.7 Anonymous Rate-Limiting Credentials

Rate limiting is part of the ARC syntax: the holder presents a credential together with a public nonce and a PRF-derived public tag, while the verifier enforces the policy by maintaining state on previously accepted $(\mathsf{nonce}, \mathsf{tag})$ pairs.

Definition 2.20 (ARC syntax)**.** Let $\mathcal{N}$ be a finite nonce set. An anonymous rate-limiting credential scheme is a tuple

$$\mathsf{ARC} = (\mathsf{Setup}, \mathsf{IssuerKeyGen}, \mathsf{Issue}, \mathsf{Show}, \mathsf{Verify})$$

where Issue outputs a credential, $\mathsf{Show}(\mathsf{cred}, \mathsf{nonce})$ outputs a presentation, and $\mathsf{Verify}(\mathsf{vk}, \mathsf{pres}, \mathsf{state})$ checks the presentation and updates verifier state.

Definition 2.21 (ARC soundness)**.** Let ARC have nonce space $\mathcal{N}$. After at most Q issuance interactions, no PPT adversary should be able to produce more than $Q|\mathcal{N}|$ accepted presentations, except with negligible probability.

Definition 2.22 (Presentation unlinkability)**.** An ARC scheme has presentation unlinkability if, for distinct fresh nonces, presentations produced from the same credential are computationally indistinguishable from presentations produced from independently issued credentials, up to the intended linkage caused by nonce reuse.

Remark 2.23 (Scope of the ARC theorems)**.** Section 4 does not prove Definition 2.21 in full. It records a conditional policy-soundness statement and a conditional presentation-unlinkability statement for the ARC construction induced by the current issuance layer.

2.8 Pseudorandom functions

Definition 2.24 (Pseudorandom function)**.** Let $\mathcal{K}, \mathcal{D}, \mathcal{T}$ be finite sets. A family

$$\mathsf{F} : \mathcal{K} \times \mathcal{D} \to \mathcal{T}$$

is pseudorandom if oracle access to $\mathsf{F}(k, \cdot)$ for uniform $k \xleftarrow{\$} \mathcal{K}$ is computationally indistinguishable from oracle access to a uniform random function from $\mathcal{D}$ to $\mathcal{T}$.

We instantiate the PRF with a Poseidon2-style permutation, but the PRF security of the keyed feed-forward/truncated family below is an explicit assumption rather than a theorem derived from the Poseidon2 permutation paper.

Definition 2.25 (Poseidon2-style permutation)**.** Let q be prime, let $d \geq 3$ satisfy $\gcd(d, q - 1) = 1$, and let $R_F, R_P \in \mathbb{N}$ with R_F even. A Poseidon2-style permutation alternates full external rounds and partial internal rounds, together with public round constants and invertible linear mixing maps over $\mathbb{F}_q$.

Definition 2.26 (Poseidon2-style PRF)**.** For key $\mathbf{k} \in \mathcal{K}_{\mathsf{key}} \subseteq \mathbb{F}_q^{\ell_{\mathsf{key}}}$ and public input $\mathbf{n} \in \mathbb{F}_q^{\ell_{\mathsf{nonce}}}$, define

$$\mathsf{F}(\mathbf{k}, \mathbf{n}) = \mathrm{Tr}_{\ell_{\mathsf{tag}}}\big(P(\mathbf{k} \| \mathbf{n}) + (\mathbf{k} \| \mathbf{n})\big),$$

where P is the Poseidon2-style permutation and $\mathrm{Tr}_{\ell_{\mathsf{tag}}}$ returns the first ℓ_{tag} lanes.

Assumption 2.27 (Concrete Poseidon2-style PRF assumption)**.** *For the concrete parameter choices fixed in Appendix E, the family*

$$\mathsf{F}(k, n) = \mathrm{Tr}_{\ell_{\mathsf{tag}}}\big(P(k \| n) + (k \| n)\big)$$

of Definition 2.26 is a secure pseudorandom function for $k \xleftarrow{\$} \mathcal{K}_{\mathsf{key}}$*. If the signed key space is changed, this assumption is made for the changed key distribution and the ARC relations must enforce that same key space.*

3 BLIND-ISSUANCE CONSTRUCTION

This section presents the committed-message issuance protocol underlying ARC-SPRUCE. The construction combines the target-shaped commitment of Section 2.2, the BB-tran rational hash of

Section 2.3, and the SmallWood NIZK-ARK of Section 2.5. The semantic message is $M = \mathbf{m}\|\mathbf{k}$; the DKLW rational-hash characteristic μ_{sig} is sampled by the issuer.

3.1 Committed-message target

The holder commits to the semantic message using

$$c = C_M M + A_s s + e.$$

The issuer samples the BB-tran characteristic and randomness

$$\mu_{\mathrm{sig}} \in R_q^{\ell_{\mu_{\mathrm{sig}}}}, \qquad x_0 \in R_q^{\ell_{x_0}}, \qquad x_1 \in R_q,$$

with $\ell_{\mu_{\mathrm{sig}}} = 1$ and $\ell_{x_0} = 2$ for the DKLW BB-tran parameter choice used here. If $B_3 - \mathbf{1}_n x_1$ is invertible, it sets

$$Z = (B_3 - \mathbf{1}_n x_1)^{-1}$$

and signs the target

$$T = B_0 + B_1 \mu_{\mathrm{sig}} + B_2 x_0 + Z + c. \tag{4}$$

The preimage signature is $\mathbf{u}$ with $\mathbf{A}\mathbf{u} = T$. This is the IntGenISIS-style shape $A\mathbf{u} = f(\mu_{\mathrm{sig}}, \chi_{\mathrm{sig}}) + C[M; s; e]$, with $C = [C_M \mid A_s \mid I]$.

3.2 Protocol relations

We record the two relations used later in the issuance and ARC layers.

Pre-sign issuance relation. The public statement is

$$x_{\mathrm{IssueBS}} = (\mathsf{pp}_{\mathsf{Com}}, c),$$

and the hidden witness is

$$w_{\mathrm{IssueBS}} = (M, s, e).$$

Define $\mathcal{R}_{\mathrm{IssueBS}}$ by the constraints

$$c = C_M M + A_s s + e, \tag{5}$$
$$M = \mathbf{m}\|\mathbf{k}, \tag{6}$$
$$\mathbf{k} \in \mathcal{K}_{\mathrm{key}}, \tag{7}$$

together with the coefficient bounds on M, s, e and any issuance policy constraints on $\mathbf{m}$. The proof only certifies a well-formed commitment opening and policy/key-space membership. It does not certify the BB-tran inverse equation or the final target computation, which are performed by the issuer after the proof is verified.

Post-issuance signature relation. For later knowledge statements, it is convenient to isolate the raw committed-message signature relation. The public statement is

$$x_{\mathrm{SigBS}} = (\mathbf{A}, B, C_M, A_s, \beta_{\mathrm{msg}}, \beta_{\mathtt{sig}}),$$

and the hidden witness is

$$w_{\mathrm{SigBS}} = (\mathbf{u}, M, s, e, \mu_{\mathrm{sig}}, x_0, x_1, Z).$$

Define $\mathcal{R}_{\mathrm{SigBS}}$ by

$$(B_3 - \mathbf{1}_n x_1) \odot Z = 1_n, \tag{8}$$
$$\mathbf{A}\mathbf{u} = B_0 + B_1 \mu_{\mathrm{sig}} + B_2 x_0 + Z + C_M M + A_s s + e, \tag{9}$$

with $M = \mathbf{m}\|\mathbf{k}$, $\mathbf{k} \in \mathcal{K}_{\mathrm{key}}$, and the prescribed coefficient bounds on $\mathbf{u}, M, s, e, \mu_{\mathrm{sig}}, x_0, x_1$. No coefficient bound on Z is imposed unless required by a concrete proof-system encoding.

3.3 Issuance protocol and raw verification

$\mathsf{Setup}(1^\lambda)$:

(1) Generate the commitment matrices C_M, A_s, the message encoding parameters, and distributions χ_s, χ_e.
(2) Run $\mathsf{pp}_{\mathsf{td}} \leftarrow \mathsf{TDRH.Setup}(1^\lambda, \mathsf{pp})$ to obtain the BB-tran hash description B, the rational-hash sampling spaces, the admissibility error bound δ, and the shortness bound $\beta_{\mathtt{sig}}$.
(3) Output

$$\mathsf{pp}_{\mathsf{BSig}} = (C_M, A_s, \mathsf{pp}_{\mathsf{td}}, \beta_{\mathsf{msg}}, \beta_{\mathtt{sig}}, \mathcal{K}_{\mathsf{key}}).$$

$\mathsf{IssuerKeyGen}(\mathsf{pp}_{\mathsf{BSig}})$:

(1) Compute $(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TDRH.TrapGen}(\mathsf{pp}_{\mathsf{td}})$.
(2) Output $\mathsf{vk} = \mathbf{A}$ and $\mathsf{sk} = \mathsf{td}$.

Issue: The issuer $\mathcal{I}$ holds sk while the holder $\mathcal{H}$ has input semantic message M and issuer key vk.

(1) The holder samples $s \leftarrow \chi_s^{k_s}$ and $e \leftarrow \chi_e^{n_c}$, computes

$$c = C_M M + A_s s + e,$$

and sends $(c, \pi_{\mathsf{Issue}})$ to the issuer, where

$$\pi_{\mathsf{Issue}} \leftarrow \mathsf{NIZK.Prove}\big(w_{\mathrm{IssueBS}} \mid \mathcal{R}_{\mathrm{IssueBS}}, x_{\mathrm{IssueBS}}\big).$$

(2) The issuer verifies π_{Issue}. If it fails, the issuer aborts.
(3) The issuer samples $\mu_{\mathrm{sig}}, x_0, x_1$ from the DKLW BB-tran domains. If $B_3 - \mathbf{1}_n x_1 \notin R_q^{\times n}$, it rejects this draw and resamples. The resulting failure probability is accounted for as the rational-hash admissibility error.
(4) The issuer computes

$$Z = (B_3 - \mathbf{1}_n x_1)^{-1}, \qquad T = B_0 + B_1 \mu_{\mathrm{sig}} + B_2 x_0 + Z + c.$$

(5) The issuer samples $\mathbf{u} \leftarrow \mathsf{TDRH.SampPre}(\mathsf{td}, T)$ and sends $(\mathbf{u}, \mu_{\mathrm{sig}}, x_0, x_1)$ to the holder.
(6) The holder checks

$$\mathbf{A}\mathbf{u} = B_0 + B_1 \mu_{\mathrm{sig}} + B_2 x_0 + Z + c \qquad \text{and} \qquad \|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}.$$

If the checks succeed, it stores the raw post-issuance witness

$$\mathsf{sig} = (\mathbf{u}, M, s, e, \mu_{\mathrm{sig}}, x_0, x_1).$$

$\mathsf{Verify}(\mathsf{vk}, M, \mathsf{sig})$: On input $\mathsf{sig} = (\mathbf{u}, M, s, e, \mu_{\mathrm{sig}}, x_0, x_1)$, verification rejects if any public bound or key-space check fails. It also rejects if $B_3 - \mathbf{1}_n x_1 \notin R_q^{\times n}$. Otherwise it computes $Z = (B_3 - \mathbf{1}_n x_1)^{-1}$ and accepts iff

$$\mathbf{A}\mathbf{u} = B_0 + B_1 \mu_{\mathrm{sig}} + B_2 x_0 + Z + C_M M + A_s s + e.$$

The output sig is a raw committed-message witness for the BB-tran signature relation. The ARC layer uses it only inside later zero-knowledge showing proofs; the current section does not identify it with a completed blind signature proof.

3.4 Correctness and issuer-view hiding

Proposition 3.1 (Correctness). *Assume correctness of the NIZK and correctness of the trapdoor sampler. Then honest execution of* Issue *outputs a raw post-issuance witness accepted by* Verify *except with negligible probability.*

Proof. The pre-sign NIZK is complete for the commitment opening. The issuer computes an admissible $\mu_{\mathrm{sig}}, x_0, x_1, Z$ and sets $T = B_0 + B_1\mu_{\mathrm{sig}} + B_2x_0 + Z + c$. Correctness of SampPre yields $\mathbf{A}\mathbf{u} = T$ with $\|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}$ except with negligible failure probability. The holder and raw verifier recompute the same equation and therefore accept. □

Proposition 3.2 (Honest-issuer issuer-view hiding). *Assume Assumption 2.2 for the commitment, zero knowledge of the pre-sign NIZK, and honest issuer sampling of $\mu_{\mathrm{sig}}, x_0, x_1$ independently of M. Then the honest issuer's view in issuance for any two semantic messages M_0, M_1 is computationally indistinguishable.*

Proof. The issuer receives c and π_{Issue} from the holder before generating the rational-hash part. By Lemma 2.3, c computationally hides the choice of M. Zero knowledge replaces π_{Issue} by a simulated proof depending only on the public statement. The values $\mu_{\mathrm{sig}}, x_0, x_1, Z$ and the computed target contribution $B_0 + B_1\mu_{\mathrm{sig}} + B_2x_0 + Z$ are generated by the honest issuer independently of M. Therefore the signed target

$$T = h_{\mu_{\mathrm{sig}},\chi_{\mathrm{sig}}}(B) + c$$

reveals no more about M than c, and the two issuer views are computationally indistinguishable. □

Remark 3.3 (No malicious-issuer blindness theorem). The preceding proposition is an honest-issuer privacy statement. A malicious-issuer blindness theorem would need a separate definition and proof, including how adversarial choices of the rational-hash part are handled.

4 FROM BLIND ISSUANCE TO ANONYMOUS RATE-LIMITING CREDENTIALS

This section lifts the committed-message issuance protocol of Section 3 to an anonymous rate-limiting credential (ARC). The semantic signed message is

$$M = \mathbf{m}\|\mathbf{k},$$

where $\mathbf{m}$ carries holder attributes and $\mathbf{k}$ is a secret PRF key. Issuance signs a target containing a commitment term for M, while showing proves signature validity and correct tag derivation

$$\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce})$$

for a public nonce nonce.

4.1 Why the signed key gives rate limiting

For a fixed credential witness, the hidden key $\mathbf{k}$ is fixed once issuance completes and is required to lie in the same key space $\mathcal{K}_{\mathrm{key}}$ used in Assumption 2.27. Consequently, for any public nonce nonce, the public tag tag $= \mathsf{F}(\mathbf{k}, \mathsf{nonce})$ is uniquely determined. If the holder reuses the same nonce with the same credential witness, the same tag must reappear. For distinct fresh nonces, Assumption 2.27 says that the resulting tags are computationally indistinguishable from outputs of a random function at those inputs. The verifier can therefore enforce a rate policy by maintaining local state on previously accepted $(\mathsf{nonce}, \mathsf{tag})$ pairs.

4.2 ARC relations

We now refine the committed-message issuance relation to the message split $M = \mathbf{m}\|\mathbf{k}$.

Issuance relation for ARC.. The public statement is

$$x_{\mathrm{IssueARC}} = (C_M, A_s, c),$$

and the hidden witness is

$$w_{\mathrm{IssueARC}} = (M, \mathbf{m}, \mathbf{k}, s, e).$$

Define $\mathcal{R}_{\mathrm{IssueARC}}$ by the constraints

$$c = C_M M + A_s s + e, \tag{10}$$
$$M = \mathbf{m}\|\mathbf{k}, \tag{11}$$
$$\mathbf{k} \in \mathcal{K}_{\mathrm{key}}, \tag{12}$$

together with coefficient bounds on M, s, e and any issuance policy constraints on $\mathbf{m}$. In the bounded-key instantiation, $\mathcal{K}_{\mathrm{key}} \subseteq \{\mathbf{k} : \|\mathbf{k}\|_\infty \le \beta_{\mathrm{msg}}\}$, so key membership can be implemented by the same bounded-membership gadget, plus any additional key-space constraints.

Showing relation. The public showing statement is

$$x_{\mathrm{ShowARC}} = (\mathsf{nonce}, \mathsf{tag}, \mathbf{A}, B, C_M, A_s, \beta_{\mathtt{msg}}, \beta_{\mathtt{sig}}),$$

and the hidden witness is

$$w_{\mathrm{ShowARC}} = (\mathbf{u}, M, \mathbf{m}, \mathbf{k}, s, e, \mu_{\mathrm{sig}}, x_0, x_1, Z).$$

Define $\mathcal{R}_{\mathrm{ShowARC}}$ by

$$(B_3 - \mathbf{1}_n x_1) \odot Z = 1_n, \tag{13}$$
$$\mathbf{A}\mathbf{u} = B_0 + B_1\mu_{\mathrm{sig}} + B_2x_0 + Z + C_M M + A_s s + e, \tag{14}$$
$$M = \mathbf{m}\|\mathbf{k}, \tag{15}$$
$$\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}), \tag{16}$$
$$\mathsf{nonce} \in \mathcal{N}, \tag{17}$$

and membership/bound checks

$$\mathbf{k} \in \mathcal{K}_{\mathrm{key}}, \qquad \|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}},$$

plus the prescribed bounds on $M, s, e, \mu_{\mathrm{sig}}, x_0, x_1$. The issuance commitment c is not public in showing; the proof instead reconstructs the same target-shaped commitment term internally as $C_M M + A_s s + e$.

4.3 ARC protocol

We define the anonymous rate-limiting credential scheme

$$\mathsf{ARC} = (\mathsf{Setup}, \mathsf{IssuerKeyGen}, \mathsf{Issue}, \mathsf{Show}, \mathsf{Verify})$$

using the committed-message issuance protocol of Section 3 and the relations above.

$\mathsf{Setup}(1^\lambda)$ **and** $\mathsf{IssuerKeyGen}(\mathsf{pp}_{\mathsf{BSig}})$**:** These are the same as in Section 3.

Issue**:** The issuer $\mathcal{I}$ holds sk while the holder $\mathcal{H}$ has attribute vector $\mathbf{m}$ and verification key vk.

(1) The holder samples a secret PRF key $\mathbf{k} \xleftarrow{\$} \mathcal{K}_{\mathrm{key}}$, forms

$$M = \mathbf{m}\|\mathbf{k},$$

samples $s \leftarrow \chi_s^{k_s}$ and $e \leftarrow \chi_e^{n_c}$, and computes

$$c = C_M M + A_s s + e.$$

It sends (c, π_{Issue}), where π_{Issue} proves $\mathcal{R}_{\text{IssueARC}}$, to the issuer.

(2) The issuer verifies the proof. If it accepts, it samples $\mu_{\text{sig}}, x_0, x_1$, resampling until $B_3 - \mathbf{1}_n x_1$ is invertible, sets

$$Z = (B_3 - \mathbf{1}_n x_1)^{-1}, \qquad T = B_0 + B_1\mu_{\text{sig}} + B_2 x_0 + Z + c,$$

samples $\mathbf{u} \leftarrow \mathsf{TDRH.SampPre}(\mathsf{td}, T)$, and sends $(\mathbf{u}, \mu_{\text{sig}}, x_0, x_1)$ to the holder.

(3) The holder checks $\mathbf{A}\mathbf{u} = T$ and $\|\mathbf{u}\|_\infty \le \beta_{\mathtt{sig}}$. If the checks succeed, it stores the credential witness

$$\mathsf{cred} = (\mathbf{u}, M, \mathbf{m}, \mathbf{k}, s, e, \mu_{\text{sig}}, x_0, x_1).$$

$\mathsf{Show}(\mathsf{cred}, \mathsf{nonce})$**:** On input a stored credential, the holder computes

$$Z = (B_3 - \mathbf{1}_n x_1)^{-1}, \qquad \mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}),$$

then generates a showing proof

$$\pi_{\text{Show}} \leftarrow \mathsf{NIZK.Prove}\big(w_{\text{ShowARC}} \mid \mathcal{R}_{\text{ShowARC}}, x_{\text{ShowARC}}\big).$$

It outputs the presentation

$$\mathsf{pres} = (\mathsf{nonce}, \mathsf{tag}, \pi_{\text{Show}}).$$

$\mathsf{Verify}(\mathsf{vk}, \mathsf{pres}, \mathsf{state})$**:** On input $\mathsf{pres} = (\mathsf{nonce}, \mathsf{tag}, \pi_{\text{Show}})$, the verifier checks π_{Show} against $\mathcal{R}_{\text{ShowARC}}$. If the proof rejects, it outputs $(0, \mathsf{state})$. Otherwise it checks whether $(\mathsf{nonce}, \mathsf{tag})$ is already present in state. If so, it rejects; if not, it inserts $(\mathsf{nonce}, \mathsf{tag})$ into state and outputs $(1, \mathsf{state})$.

4.4 Security decomposition

This construction does not prove Definition 2.21 in full. Instead, it records the policy-soundness and presentation-unlinkability consequences obtained from the present issuance layer, the showing proof, and the concrete PRF assumption.

Correctness. If issuance completes successfully, then the holder stores a witness satisfying

$$\mathbf{A}\mathbf{u} = B_0 + B_1\mu_{\text{sig}} + B_2 x_0 + Z + C_M M + A_s s + e, \qquad (B_3 - \mathbf{1}_n x_1) \odot Z = 1_n.$$

For any nonce, the holder can therefore satisfy the showing relation, and completeness of the NIZK makes the verifier accept unless the local rate-limiting state rejects the pair $(\mathsf{nonce}, \mathsf{tag})$.

Proposition 4.1 (Conditional policy soundness)**.** *Assume the showing NIZK is knowledge sound, the interactive issuance layer is one-more unforgeable for the raw committed-message credential witnesses extracted by $\mathcal{R}_{\text{ShowARC}}$, and the PRF evaluation is deterministic and correct. Then after at most Q issuance interactions, any PPT adversary can cause more than $Q|\mathcal{N}|$ state-accepted presentations only with negligible probability.*

Proof. Knowledge soundness of the showing proof yields a witness satisfying $\mathcal{R}_{\text{ShowARC}}$. In particular, the witness contains a semantic message $M = \mathbf{m}\|\mathbf{k}$ and a valid preimage equation for the target containing the commitment term $C_M M + A_s s + e$. By the assumed one-more unforgeability of the issuance layer, at most Q distinct valid raw credential witnesses can be produced after Q issuance interactions, except with negligible probability. For each fixed credential witness and each fixed nonce, PRF correctness fixes a unique tag. Therefore a stateful verifier that rejects repeated $(\mathsf{nonce}, \mathsf{tag})$ pairs can accept at most one presentation per nonce per valid credential witness. This gives the bound $Q|\mathcal{N}|$; accidental tag collisions only reduce the number of accepted pairs. □

Proposition 4.2 (Conditional presentation unlinkability)**.** *Assume honest-issuer issuer-view hiding of the issuance layer, zero knowledge of the showing proof, and Assumption 2.27 for keys sampled from $\mathcal{K}_{\text{key}}$. Then, for distinct fresh nonces, presentations produced from the same credential are computationally indistinguishable from presentations produced from independently issued credentials, except for the intended linkage caused by nonce reuse.*

Proof. Honest-issuer issuer-view hiding prevents the honest issuance transcript from revealing the committed message. Zero knowledge of the showing proof hides the witness $(\mathbf{u}, M, \mathbf{m}, \mathbf{k}, s, e, \mu_{\text{sig}}, x_0, x_1, Z)$ during presentation. For distinct fresh nonces, Assumption 2.27 makes the values $\mathsf{F}(\mathbf{k}, \mathsf{nonce})$ computationally indistinguishable from random-function outputs at those inputs, hence tags do not reveal whether two presentations used the same hidden key. The only intended exception is nonce reuse, where determinism of the PRF causes the tag to repeat and rate limiting deliberately creates linkage. □

Attribute privacy and selective disclosure. In the core ARC relation of this paper, $\mathbf{m}$ remains hidden during showing. Predicate proofs or selective disclosure can be added by extending $\mathcal{R}_{\text{ShowARC}}$ with further constraints on $\mathbf{m}$, without changing the issuance protocol or the signed-message format.

5 SMALLWOOD PROGRAMMING MODEL

This section gives the SmallWood formulation of the ARC-SPRUCE NIZKs. The prover commits to packed witness rows, represented as low-degree row polynomials over a packing set Ω and masked on a pairwise-disjoint hiding set Ω'. The verifier then checks PACS-style constraints on those committed rows. We record here the semantic statement layers and the exact coefficient-domain consequences needed by the security arguments; concrete row inventories, gadget-level realizations, and proof-stack engineering are deferred to Appendix C.

5.1 Proof interface

Fix a packing set $\Omega = \{\omega_1, \ldots, \omega_s\} \subset \mathbb{F}_q$ and a disjoint hiding set $\Omega' = \{\omega'_1, \ldots, \omega'_\ell\} \subset \mathbb{F}_q \setminus \Omega$. A compiled witness is a matrix

$$W \in \mathbb{F}_q^{n \times s}.$$

For each row i, the prover interpolates a polynomial $P_i \in \mathbb{F}_q[X]$ of degree $< s + \ell$ such that $P_i(\omega_j) = W_{i,j}$ for all $\omega_j \in \Omega$, while the values on Ω' are random masks. The committed witness is therefore a family of row polynomials $P = (P_1, \ldots, P_n)$.

SmallWood proves families of constraints over these rows. Parallel constraints are slotwise identities; aggregated constraints are domain-summed identities. In ARC-SPRUCE, the target-shaped commitment equation and the signature equation are linear coefficient-wise constraints; the inverse-witness equation $(B_3 - \mathbf{1}_n x_1) \odot Z = 1_n$ is bilinear; and the PRF checks contain the nonlinear Poseidon2-style S-box constraints. Range, carry, packing-consistency, and norm gadgets may additionally use aggregated checks. The whole showing relation is therefore not linear, but its only rational-hash nonlinearity is the bilinear inverse check.

5.2 Issuance as a SmallWood statement

The public issuance statement is

$$x^{\mathrm{SW}}_{\mathrm{Issue}} = (C_M, A_s, c),$$

and the hidden witness is

$$w^{\mathrm{SW}}_{\mathrm{Issue}} = (M, \mathbf{m}, \mathbf{k}, s, e).$$

The committed witness rows encode the semantic message M, the message/key split $(\mathbf{m}, \mathbf{k})$, the commitment randomness $s \in R^{k_s}$, and the error $e \in R^{n_c}$. Semantically, the proof certifies:

$$\begin{aligned} c &= C_M M + A_s s + e, \\ M &= \mathbf{m}\|\mathbf{k}, \\ \mathbf{k} &\in \mathcal{K}_{\mathrm{key}}, \end{aligned}$$

together with coefficient bounds on M, s, e and any issuance policy constraints on $\mathbf{m}$. The issuance proof carries no rational-hash inverse constraint and no target computation constraint. The issuer samples the rational-hash values only after this proof verifies.

5.3 Showing as a SmallWood statement

The public showing statement is

$$x^{\mathrm{SW}}_{\mathrm{Show}} = (\mathsf{nonce}, \mathsf{tag}, \mathbf{A}, B, C_M, A_s, \beta_{\mathtt{msg}}, \beta_{\mathtt{sig}}),$$

with verifier state handled outside the proof. The issuance commitment c is not included in the public showing statement. The hidden witness is

$$w^{\mathrm{SW}}_{\mathrm{Show}} = (\mathbf{u}, M, \mathbf{m}, \mathbf{k}, s, e, \mu_{\mathrm{sig}}, x_0, x_1, Z).$$

The committed rows encode the short signature witness $\mathbf{u}$, the semantic message and split $(M, \mathbf{m}, \mathbf{k})$, the commitment opening (s, e), the DKLW rational-hash characteristic and randomness $(\mu_{\mathrm{sig}}, x_0, x_1)$, the inverse witness Z, and the PRF companion rows.

Semantically, the proof certifies the following coefficient-domain statement:

$$(B_3 - \mathbf{1}_n x_1) \odot Z = 1_n, \tag{18}$$

$$\mathbf{Au} = B_0 + B_1\mu_{\mathrm{sig}} + B_2 x_0 + Z + C_M M + A_s s + e, \tag{19}$$

$$M = \mathbf{m}\|\mathbf{k}, \tag{20}$$

$$\mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}), \tag{21}$$

together with key-space membership and the coefficient bounds on all bounded witness components. Eliminating Z yields

$$(B_3 - \mathbf{1}_n x_1) \odot \big(\mathbf{Au} - B_0 - B_1\mu_{\mathrm{sig}} - B_2 x_0 - C_M M - A_s s - e\big) = 1_n, \tag{22}$$

which is the coefficient-domain residual used in security arguments.

5.4 Replay / transform implementation

Some implementations check the showing relation in a transform or replay basis. Let

$$\mathcal{F}_{\mathrm{rep}} : R_q \to \mathcal{T}$$

be the public replay map, with block projections $\mathcal{F}_{\mathrm{rep},b}$. The prover then carries replay rows for the images of

$$\mathbf{Au}, \quad \mu_{\mathrm{sig}}, \quad x_0, \quad x_1, \quad C_M M + A_s s + e, \quad Z,$$

and exact bridge constraints tie these rows to the coefficient-domain rows. A replay residual has the form

$$(B^{\Theta}_{3,b} - \widehat{x}_{1,b}) \odot \big(\widehat{T}_b - B^{\Theta}_{0,b} - B^{\Theta}_{1,b} \odot \widehat{\mu}_{\mathrm{sig},b} - B^{\Theta}_{2,b} \odot \widehat{x}_{0,b} - \widehat{c}_b\big) = \mathcal{F}_{\mathrm{rep},b}(1), \tag{23}$$

where $\widehat{T}_b$ is the replay image of $\mathbf{Au}$, $\widehat{c}_b$ is the replay image of $C_M M + A_s s + e$, and $B^{\Theta}_{i,b}$ denotes the public replay image of the relevant BB-tran coefficient block.

Assumption 5.1 (Exact replay map)**.** *The public replay map $\mathcal{F}_{\mathrm{rep}}$ used by the showing implementation is additive, multiplicative on all products used in the compiled residual, and injective on the coefficient-domain values that appear in the showing relation. If a concrete replay representation is not injective, the compiled relation must include explicit linking constraints sufficient to recover the same coefficient-domain identities.*

Theorem 5.2 (Replay-space certification implies the coefficient-domain committed-message identity)**.** *Assume Assumption 5.1. Assume further that the compiled showing relation enforces the exact bridges above and the replay residual* (23)*. Then the extracted coefficient-domain witness satisfies*

$$(B_3 - \mathbf{1}_n x_1) \odot \big(\mathbf{Au} - B_0 - B_1\mu_{\mathrm{sig}} - B_2 x_0 - C_M M - A_s s - e\big) = 1_n$$

in R_q.

Proof. Substitute the exact bridge equalities into the replay residual. Additivity and multiplicativity of $\mathcal{F}_{\mathrm{rep}}$ transform the replay residual into the replay image of the coefficient-domain residual

$$(B_3 - \mathbf{1}_n x_1) \odot \big(\mathbf{Au} - B_0 - B_1\mu_{\mathrm{sig}} - B_2 x_0 - C_M M - A_s s - e\big) - 1_n.$$

Injectivity then forces this coefficient-domain residual itself to vanish, which is exactly the displayed identity. □

Corollary 5.3 (Abstract committed-message witness consequence)**.** *Under the assumptions of Theorem 5.2, the extracted witness additionally satisfies*

$$(B_3 - \mathbf{1}_n x_1) \odot Z = 1_n, \qquad \mathbf{Au} = h_{\mu_{\mathrm{sig}},\chi_{\mathrm{sig}}}(B) + C_M M + A_s s + e, \qquad \mathsf{tag} = \mathsf{F}(\mathbf{k}, \mathsf{nonce}).$$

Proof. The inverse equation $(B_3 - \mathbf{1}_n x_1) \odot Z = 1_n$ is part of the compiled relation. By Theorem 5.2, the bracketed residual multiplied by $B_3 - \mathbf{1}_n x_1$ is 1_n. Combining this with the inverse equation identifies the bracketed term with Z, hence the displayed signature equation. The PRF equation is already part of the showing relation. □

5.5 PRF companion relation

The public tag is tied to the same hidden key carried in $M = \mathbf{m}\|\mathbf{k}$ through a companion PRF relation. Selector constraints bind the PRF key lanes to the key rows $\mathbf{k}$ and enforce $\mathbf{k} \in \mathcal{K}_{\mathrm{key}}$; checkpoint constraints certify the nonlinear Poseidon2-style S-box steps; linear layers and round constants are replayed from public parameters; and the final feed-forward and truncation constraints bind the public pair $(\mathsf{nonce}, \mathsf{tag})$.

A crucial invariant is that the PRF key rows are not separate from the signature/message rows: the compiled showing statement is a single committed witness family. If auxiliary alias rows are introduced for engineering convenience, they must be tied back to the original key rows by explicit copy constraints. This prevents a prover from using one hidden key for the committed-message signature relation and another hidden key for the PRF relation.

5.6 Statement strength

The SmallWood proof hides a family of row polynomials, not merely a tuple of scalars. Random masks on Ω' statistically hide unopened row values, while the DECS/LVCS/PCS stack reveals only the evaluations needed to check the PACS constraints. Consequently, the verifier learns only the public statement: during issuance this is c together with public parameters, and during showing it is $(\mathsf{nonce}, \mathsf{tag})$ together with the public verification context. The verifier does not see c, M, s, e, μ_{sig}, x_0, x_1, Z, or $\mathbf{u}$ in the showing protocol.

Corollary 5.4 (Consequence of an accepting showing proof)**.** *If the showing NIZK is knowledge sound for the compiled showing relation, then every accepting proof yields a coefficient-domain witness satisfying* (18)–(21) *and the stated bounds.*

6 PARAMETERS AND INTEGRATION

This section explains how parameters interact across the ARC-SPRUCE stack: the target-shaped commitment, the BB-tran rational hash, the lattice preimage sampler, SmallWood, and the PRF. Detailed parameter discussion and benchmark context are deferred to Appendix D.

6.1 Base ring and primary profile

All components share the same base ring R_q and modulus q: the commitment matrices, signature map $\mathbf{A}$, and hash key B are defined over R_q; SmallWood constraints are evaluated over $\mathbb{F}_q$; and the PRF is instantiated over $\mathbb{F}_q$ so that its constraints are native to the proof system. The primary parameter profile is

$$N = 512, \qquad q = 1054721, \qquad \ell_M = 1, \qquad k_s = 2, \qquad n_c = 1, \qquad \beta_{\text{msg}} = 8.$$

The condition $\ell_M = 1$ is an encoding assumption for $M = \mathbf{m}\|\mathbf{k}$. If the semantic message does not fit into one ring polynomial, ℓ_M must be increased and the commitment dimensions, binding length, MLWE/MSIS estimates, and SmallWood row inventories must be recomputed.

6.2 Commitment parameters

The commitment term is

$$c = C_M M + A_s s + e \in R_q^{n_c},$$

where

$$C_M \in R_q^{n_c \times \ell_M}, \qquad A_s \in R_q^{n_c \times k_s}, \qquad s \in R^{k_s}, \qquad e \in R^{n_c}.$$

For the primary profile,

$$C_M \in R_q^{1\times 1}, \qquad A_s \in R_q^{1\times 2}, \qquad s \in R^2, \qquad e \in R.$$

Thus the commitment opening contains three small ring polynomials beyond the semantic message: two secret polynomials in s and one error polynomial e. The coefficients of s and e are sampled uniformly from $[-8, 8]$. For estimator reporting, this distribution has coefficient standard deviation

$$\sigma_B = \sqrt{B(B+1)/3} = \sqrt{24} \qquad \text{for } B = 8.$$

The proof bound $B = 8$, the estimator standard deviation σ_B, the signature preimage bound $\beta_{\mathtt{sig}}$, and the message/key bound β_{msg} are distinct quantities; the first parameter profile sets $B = \beta_{\text{msg}} = 8$ for the opened commitment coordinates.

The binding matrix is

$$[C_M \mid A_s \mid I_{n_c}],$$

so the binding input length is

$$L_{\text{bind}} = \ell_M + k_s + n_c = 1 + 2 + 1 = 4.$$

For double openings with all opened coefficients bounded by $\beta_{\text{msg}} = 8$, the coefficient-infinity bound on the difference vector is

$$\beta_{\text{bind}} = 2\beta_{\text{msg}} = 16.$$

The hiding claim is computational under Assumption 2.2; there is no information-theoretic commitment-hiding claim.

6.3 BB-tran rational-hash parameters

The DKLW BB-tran part uses issuer-sampled values

$$\ell_{\mu_{\text{sig}}} = 1, \qquad \ell_{x_0} = 2, \qquad x_1 \in R_q.$$

The target contribution is

$$h_{\mu_{\text{sig}},\chi_{\text{sig}}}(B) = B_0 + B_1\mu_{\text{sig}} + B_2 x_0 + Z, \qquad (B_3 - \mathbf{1}_n x_1) \odot Z = 1_n.$$

The complete signed target is

$$T = B_0 + B_1\mu_{\text{sig}} + B_2 x_0 + Z + C_M M + A_s s + e.$$

Honest target derivation restarts when $B_3 - \mathbf{1}_n x_1 \notin R_q^{\times n}$. The restart probability must be negligible or explicitly carried as the rational-hash admissibility error δ. The semantic message M never appears in the B_1 rational-hash term.

Non-blind reduction parameters. The non-blind reduction imported from DKLW is parameterized by a norm bound β_{DKLW}, an inadmissibility bound δ, and the predicate/linear-independence side conditions of the BB-tran family. The public verifier bound $\beta_{\mathtt{sig}}$ should dominate the relevant DKLW norm bound under the chosen norm comparison. The IntGenISIS lifting is cited only under its stated side conditions and is not a complete proof of the interactive ARC issuance layer.

6.4 Sampler parameters

The trapdoor sampler must output short signatures $\mathbf{u}$ with overwhelming probability and with a distribution close to the required lattice-coset distribution for the imported signature theorem. This requires choosing the ring degree and modulus, trapdoor quality, Gaussian widths, and an acceptance predicate that yields the public verification bound $\beta_{\mathtt{sig}}$. Appendix B describes the NTRU/Antrag-based candidate used for tuning. The commitment design does not change the sampler/trapdoor admissibility interface.

6.5 SmallWood row arithmetic

The pre-sign proof checks the commitment opening, message encoding, key-space membership, bounds, and policy constraints. The showing proof checks the signature equation, the bilinear inverse equation, the hidden commitment term, the message encoding, and the PRF checkpoint constraints.

Ignoring PRF internals and auxiliary range rows, the primary profile has the following showing witness inventory:

$$\mathbf{u}: 2, \quad M: 1, \quad s: 2, \quad e: 1, \quad \mu_{\text{sig}}: 1, \quad x_0: 2, \quad x_1: 1, \quad Z: 1.$$

This is a total of

$$2 + 1 + 2 + 1 + 1 + 2 + 1 + 1 = 11$$

ring polynomials. With SmallWood packing $s_{\mathsf{SW}} = 16$, one ring polynomial contributes

$$N/s_{\mathsf{SW}} = 512/16 = 32$$

witness rows. Therefore the non-PRF showing witness inventory contributes

$$11 \cdot 32 = 352$$

SmallWood rows before PRF auxiliary rows. The pre-sign opening inventory is (M, s, e), namely

$$1 + 2 + 1 = 4$$

ring polynomials, contributing $4 \cdot 32 = 128$ rows at the same packing. These are row-count arithmetic statements only; final proof sizes require running the SmallWood optimizer with the actual row inventory, constraint degrees, Fiat–Shamir parameters, and PRF checkpoint schedule.

6.6 Secondary compact profile

A smaller-ring candidate is

$$N = 256, \qquad q = 1054721, \qquad \ell_M = 1, \qquad k_s = 4, \qquad n_c = 1, \qquad B = 8.$$

Its MLWE hiding estimate is approximately 194.4 bits for the estimator configuration used in the parameter notes. The corresponding estimator dimensions are

$$m_{\mathsf{LWE}} = Nn_c = 256, \qquad m_{\mathsf{SIS}} = N(\ell_M + k_s + n_c) = 256(1+4+1) = 1536.$$

Its non-PRF showing witness inventory is

$$\mathbf{u} : 2, \quad M : 1, \quad s : 4, \quad e : 1, \quad \mu_{\mathsf{sig}} : 1, \quad x_0 : 2, \quad x_1 : 1, \quad Z : 1,$$

for a total of 13 ring polynomials. With $s_{\mathsf{SW}} = 16$, this contributes

$$13 \cdot (256/16) = 208$$

SmallWood rows before PRF auxiliary rows. A binding estimate must be inserted before this profile is claimed as a full concrete commitment-security profile.

6.7 PRF parameters

The PRF must be secure at the target security level, efficiently representable in constraints, and compatible with $\mathbb{F}_q$. We use a Poseidon2-style permutation with feed-forward and truncation. Its parameters include the state width t, round counts (R_F, R_P), MDS matrices, round constants, S-box exponent d, truncation length ℓ_{tag}, and key space $\mathcal{K}_{\mathsf{key}}$. We require $\ell_{\mathsf{tag}} \geq \lceil \lambda/\log_2(q) \rceil$ as an output-length condition; pseudorandomness of the resulting tags is exactly Assumption 2.27 for keys sampled from $\mathcal{K}_{\mathsf{key}}$. The output-length condition alone is not a proof of PRF security.

A ADDITIONAL PRELIMINARIES

A.1 Security assumptions for lattice signatures

Fix a public left factor F, hint space H, target space G, predicate P, and query bound Q.

- **Hint-vSIS.** The adversary chooses a query set $Q \subseteq H$ and a fresh target function $g^\star \in G \setminus Q$ before seeing the sampled matrix A. It then receives short preimages of the queried hints $q(A)$ under $F(A)$ for every $q \in Q$, and must output a non-zero short vector $u^\star$ such that $F(A)u^\star = g^\star(A)$. The instance is restricted by the predicate condition $P(F \cup Q \cup (\{g^\star\} \setminus \{0\})) = 1$.
- **s-Hint-vSIS.** This is the strong variant of the same experiment. The adversary still chooses the hint set Q before seeing A, but it is allowed to choose the fresh target $g^\star$ only after seeing A and the hinted preimages.
- **s-\$-Hint-vSIS.** This is the strong random-hinted variant. It is the same as s-Hint-vSIS, except that the query set Q is sampled uniformly at random from H rather than chosen by the adversary.

Now fix a keyed evaluation family $f : K \times M \times X \to R_q^n$.

- **GenISIS$_f$.** The challenger samples a public key $\kappa \leftarrow K$ and a uniform matrix $A \xleftarrow{\$} R_q^{n\times m}$. It provides a preimage oracle that, on input $\mu \in M$, samples $\chi \leftarrow X$, sets $t = f(\kappa, \mu, \chi)$, returns a short preimage u such that $Au = t$, and records the tuple (u, μ, χ). The adversary wins if it outputs a fresh tuple $(u^\star, \mu^\star, \chi^\star)$ not previously returned by the oracle such that $Au^\star = f(\kappa, \mu^\star, \chi^\star)$ and $0 < \|u^\star\| \leq \beta$.
- **IntGenISIS$_f$.** This is the interactive version used for committed messages. The challenger samples an additional public matrix C. On a bounded opening (m, r), the oracle samples the rational-hash inputs (μ, χ), sets

$$t = f(\kappa, \mu, \chi) + C[m; r],$$

  and returns a short preimage. The DKLW experiment records the full tuple (u, μ, χ, m, r) and asks the adversary to produce a fresh valid tuple.

The DKLW IntGenISIS$_f$-to-GenISIS$_f$ reduction is used only under the parameter conditions of their lifting theorem, including its ring, modulus, smoothing, dimension, and bound-loss requirements [5, Theorem 6].

A.2 Vanishing polynomials for membership constraints

Membership constraints are expressed using vanishing polynomials over $\mathbb{F}_q$ with signed lifting.

Definition A.1 (Vanishing polynomial for $[-\mathsf{BoundB}, \mathsf{BoundB}]$). Let $\mathsf{BoundB} \in \mathbb{Z}_{>0}$. Define

$$P_{\mathsf{BoundB}}(X) \;=\; \prod_{i=-\mathsf{BoundB}}^{\mathsf{BoundB}} \big(X - \langle i \rangle_q\big) \;\in\; \mathbb{F}_q[X].$$

For $x \in \mathbb{F}_q$, we have $x \in [-\mathsf{BoundB}, \mathsf{BoundB}]$ via signed lifting iff $P_{\mathsf{BoundB}}(x) = 0$ in $\mathbb{F}_q$.

Definition A.2 (Vanishing polynomial for $\{-1, 0, 1\}$). Define $P_1(X) = (X + 1)X(X - 1) \in \mathbb{F}_q[X]$. Then $P_1(x) = 0$ iff $x \in \{-1, 0, 1\}$.

B GAUSSIAN PREIMAGE SAMPLING

This appendix describes the Gaussian preimage-sampling interface used by the issuer to sign a target $t \in R_q^n$ by producing a short preimage $u \in R^m$ such that $Au = t$ in R_q^n. The security import in

Section 2.4 requires the admissible trapdoor-suite interface of Definition 2.6. The concrete implementation candidate described here follows the vSIS–BB-tran setting [5] and uses an NTRU trapdoor generated via Antrag together with a hybrid two-plane sampler [6]. Antrag is used here as a trapdoor-generation and quality-tuning source, not as a standalone proof of the full DKLW admissible-suite interface.

B.1 The preimage sampling problem

Given a public linear map $A \in R_q^{n\times m}$ and a target $t \in R_q^n$, the signer samples

$$u \leftarrow \mathsf{SampPre}(\mathsf{td}_A, t) \qquad \text{such that} \qquad Au \equiv t \pmod q$$

and u is short (according to a public bound $\beta_{\mathtt{sig}}$). Equivalently, u lies in the coset lattice

$$\Lambda_\perp^t(A) \coloneqq \{\, x \in R^m : Ax \equiv t \ (\mathrm{mod}\ q) \,\}.$$

For compatibility with the DKLW import, the sampler layer must provide: (i) correctness ($Au = t$ and the norm test fails only with negligible probability), (ii) a preimage distribution close enough to the required lattice-coset Gaussian or an explicitly assumed substitute distributional guarantee, and (iii) a tail bound that justifies the public verification bound $\beta_{\mathtt{sig}}$ in the coefficient ℓ_∞-norm used by the ARC relations.

B.2 NTRU trapdoor form and quality

We work with the rank-2 NTRU module over the cyclotomic ring $R = \mathbb{Z}[X]/(X^N + 1)$ and its quotient R_q. An NTRU public key has the form $h = gf^{-1} \bmod q$ for secret $f, g \in R$. The corresponding module is

$$\mathsf{L}_{\mathsf{NTRU}}(h) \;=\; \{(u, v) \in R^2 : uh - v \equiv 0 \ (\mathrm{mod} q)\}.$$

A trapdoor basis of $\mathsf{L}_{\mathsf{NTRU}}(h)$ is

$$B_{f,g} \;=\; \begin{pmatrix} f & F \\ g & G \end{pmatrix}, \qquad \text{with} \qquad fG - gF \;=\; q,$$

whose columns are short and satisfy $g/f \equiv G/F \equiv h \pmod q$. Following [6], we measure trapdoor quality via per-slot energies under the canonical embeddings φ_i, requiring a uniform window around q. Concretely, for a quality parameter $\alpha > 1$ we target

$$\frac{q}{\alpha^2} \;\le\; |\varphi_i(f)|^2 + |\varphi_i(g)|^2 \;\le\; \alpha^2 q \qquad \text{for all slots } i.$$

Smaller α corresponds to tighter trapdoors and, for fixed N, q, to shorter signatures for the same security.

B.3 Sampler definition and tuning

Our candidate sampler is a ring-based hybrid two-plane sampler in the evaluation domain. The sampler's main tuning parameters are:

- modulus q and ring degree N,
- trapdoor quality α,
- acceptance radius R_{acc} controlling rejection and norm tails,
- per-slot widths $\sigma_k[i]$ for the two planes $k \in \{1, 2\}$.

The concrete calibration formulas are implementation/tuning rules for the candidate sampler. The paper's signature-layer theorem uses them only through the admissible-suite properties in Definition 2.6: correctness, the required distributional hiding guarantee, and a tail bound implying $\|u\|_\infty \le \beta_{\mathtt{sig}}$ except with negligible probability.

Per-slot calibration. Let $\widetilde{b}_k[i]$ denote the per-slot Gram–Schmidt lengths for plane k at slot i. A standard calibration equalizes acceptance probability across slots:

$$\sigma_k[i] \;=\; \sqrt{\frac{\alpha^2 q\, R_{\mathrm{acc}}^2}{\|\widetilde{b}_k[i]\|^2} \;-\; R_{\mathrm{acc}}^2} \qquad (k \in \{1, 2\}).$$

This prevents weak slots and yields consistent output variance.

Smoothing threshold. To obtain the sampler distributional guarantees required by the trapdoor interface, we enforce a floor $\sigma_{\min}$ such that $\sigma_k[i] \ge \sigma_{\min}$ for all k, i, where $\sigma_{\min}$ exceeds the smoothing parameter of the relevant lattice cosets. In practice, we apply a global scale factor to raise all $\sigma_k[i]$ above the theoretical floor with slack.

B.4 Annulus sampling (Antrag key generation)

Antrag replaces trial-and-error trapdoor generation with direct annulus sampling in the Fourier domain [6]. For each conjugate pair (there are $N/2$ independent slots), the key generator samples $(|\varphi_i(f)|, |\varphi_i(g)|)$ uniformly in an annulus arc $A^+(r, R)$ with

$$\sqrt{q}/\alpha \;<\; r \;<\; R \;<\; \alpha\sqrt{q},$$

chooses random phases, inverse-embeds to the coefficient domain, and rounds to obtain $(f, g) \in R^2$. By selecting a "middle" annulus window (see [6]) one ensures that the rounded trapdoor satisfies the desired α-window with high probability and with modest repetition.

B.5 Hybrid two-plane sampler (details)

We sketch the two-plane hybrid sampler in the notation of [6]. Let td_A encode an NTRU trapdoor for A and let $t \in R_q^n$ be the target. Write $\langle\cdot\rangle_q$ for signed coefficient lift.

Algorithm 1 $\mathsf{SampPre}(\mathsf{td}_A, t)$: two-plane hybrid sampler (sketch)

Lift t to signed coefficients and set $c \leftarrow \langle t\rangle_q \in (-\frac{q}{2}, \frac{q}{2}]^n$.
Using td_A, compute per-slot widths $\sigma_k[i]$ (two planes) calibrated to trapdoor quality.
repeat
    Sample slot-wise Gaussian pairs on the two planes, reconstruct a coefficient-domain candidate u.
    Accept iff the norm predicate holds for u (e.g., ℓ_2 and/or ℓ_∞ bound).
until accepted
return u.

Public relations. Downstream verification and proofs use only the public relation $Au = t$ and shortness of u bounded by $\beta_{\mathtt{sig}}$. The value of $\beta_{\mathtt{sig}}$ must be derived from the sampler's accepted-output tail bound and converted to the coefficient ℓ_∞-norm convention used in the ARC relations.

C SMALLWOOD PROOF-STACK DETAILS

This appendix records schematic compiled surfaces used by the issuance and showing statements. It is descriptive: the semantic statements are those of Section 5, and final proof-size claims require the complete row inventory, constraint degrees, and soundness calculation for the instantiated relation.

C.1 Row inventories

Issuance rows. The issuance proof commits rows encoding:

$$M, \quad \mathbf{m}, \quad \mathbf{k}, \quad s, \quad e.$$

The active relations are

$$\begin{aligned} c &= C_M M + A_s s + e, \\ M &= \mathbf{m} \| \mathbf{k}, \\ \mathbf{k} &\in \mathcal{K}_{\mathrm{key}}, \end{aligned}$$

plus the prescribed coefficient bounds and issuance policy constraints. In the bounded-key instantiation, key membership is implemented with the same signed-lift bounded-membership convention as the attribute rows, together with any additional key-space equations.

Showing rows. The showing proof commits rows encoding:

$$\mathbf{u}, \quad M, \quad \mathbf{m}, \quad \mathbf{k}, \quad s, \quad e, \quad \mu_{\mathrm{sig}}, \quad x_0, \quad x_1, \quad Z,$$

plus rows for the signature shortness gadget and the PRF companion relation. The active algebraic relations are

$$\begin{aligned} (B_3 - \mathbf{1}_n x_1) \odot Z &= 1_n, \\ \mathbf{A}\mathbf{u} &= B_0 + B_1 \mu_{\mathrm{sig}} + B_2 x_0 + Z + C_M M + A_s s + e, \\ M &= \mathbf{m} \| \mathbf{k}, \\ \mathrm{tag} &= \mathsf{F}(\mathbf{k}, \mathrm{nonce}). \end{aligned}$$

Equivalently, the showing proof implies the eliminated coefficient-domain relation

$$(B_3 - \mathbf{1}_n x_1) \odot \big(\mathbf{A}\mathbf{u} - B_0 - B_1 \mu_{\mathrm{sig}} - B_2 x_0 - C_M M - A_s s - e\big) = 1_n.$$

C.2 Commitment constraints

The commitment opening constraint is linear over R_q and is compiled coefficient-wise into SmallWood rows:

$$c = C_M M + A_s s + e.$$

The block I_{n_c} multiplying e is public and requires no invertibility check. Binding of two distinct openings is external to the proof system and follows from the Module-SIS assumption for $[C_M \mid A_s \mid I_{n_c}]$.

C.3 BB-tran constraints

The BB-tran constraints are expressed in witness form. Showing uses the inverse equation

$$(B_3 - \mathbf{1}_n x_1) \odot Z = 1_n$$

and the signature surface

$$\mathbf{A}\mathbf{u} = B_0 + B_1 \mu_{\mathrm{sig}} + B_2 x_0 + Z + C_M M + A_s s + e.$$

The first equation is bilinear in (x_1, Z). The second equation is linear in the committed witness rows once Z is carried explicitly.

C.4 PRF companion constraints

The PRF companion relation binds the public pair $(\mathsf{nonce}, \mathsf{tag})$ to the hidden key $\mathbf{k}$ carried inside M. The proof commits key-lane rows, selected S-box checkpoint rows, and any auxiliary rows needed to replay the Poseidon2-style linear layers between checkpoints. Selector constraints tie the key-lane rows to the signed key coordinates and enforce $\mathbf{k} \in \mathcal{K}_{\mathrm{key}}$; final feed-forward and truncation constraints tie the public tag to the resulting PRF output. Security of this keyed construction is the explicit PRF assumption of Assumption 2.27.

C.5 Single committed witness

All rows in one proof live under one SmallWood commitment. This ensures that the prover cannot use one hidden message for the commitment term, a different rational-hash characteristic for the signature relation, and a third key for the PRF. In the showing proof, the same hidden key $\mathbf{k}$ is used simultaneously in the semantic message encoding and the tag relation.

D EXTENDED PARAMETER DISCUSSION

This appendix records parameter checks for the committed-message issuance relation, the compact MLWE-hiding commitment, and the SmallWood statement inventory.

D.1 Admissibility checklist

A parameter tuple is *admissible* for ARC-SPRUCE if it simultaneously satisfies:

(1) **Bound vs. modulus:** $\beta_{\mathtt{msg}} \ll q$ so membership checks on $[-\beta_{\mathtt{msg}}, \beta_{\mathtt{msg}}]$ do not wrap modulo q.
(2) **Commitment hiding:** the distribution $(A_s, A_s s + e)$ satisfies the Module-LWE hiding assumption for the selected $(R, q, n_c, k_s, \chi_s, \chi_e)$.
(3) **Commitment binding:** double openings are covered by the Module-SIS binding assumption for $[C_M \mid A_s \mid I_{n_c}]$, input length $L_{\mathrm{bind}} = \ell_M + k_s + n_c$, and bound β_{bind}.
(4) **Sampler interface:** the trapdoor/preimage layer satisfies Definition 2.6, including the public ℓ_∞ shortness bound $\beta_{\mathtt{sig}}$.
(5) **Rational-hash guard:** denominator invertibility failure for $B_3 - \mathbf{1}_n x_1$ is negligible or explicitly carried as the rational-hash admissibility error δ.
(6) **SmallWood degree/soundness:** the DECS/LVCS/PCS/PIOP degree bounds and soundness parameters are computed for the actual issuance and showing relations.
(7) **PRF key space and tag length:** the relation enforces $\mathbf{k} \in \mathcal{K}_{\mathrm{key}}$, the PRF assumption is made for this same key space, and ℓ_{tag} is large enough for the intended tag length.

D.2 Sampler budgets

Let $\eta_\varepsilon(\Lambda)$ denote the smoothing parameter of a lattice Λ. A sufficient condition for the preimage distribution used by the signature import is to choose Gaussian widths above the relevant smoothing floor, with slack. For the ambient lattice $\mathbb{Z}^{2N}$ one may use the classical bound

$$\eta_\varepsilon(\mathbb{Z}^{2N}) = \sqrt{\ln\big(4N(1 + 2^\lambda)\big)/\pi}.$$

In the sampler tuning, trapdoor quality α and acceptance parameters are chosen so that the output shortness bound $\beta_{\mathtt{sig}}$ holds with overwhelming probability.

D.3 Primary lattice and commitment profile

Table 1 records the primary profile used for the row-inventory discussion.

Knob / Result	Symbol	Value
Ring degree	N	512
Modulus	q	1054721
Semantic message length	ℓ_M	1
Commitment secret length	k_s	2
Commitment target length	n_c	1
Message/randomness bound	β_{msg}	8
MLWE sample dimension	m_{LWE}	$Nn_c = 512$
MSIS input dimension	m_{SIS}	$N(\ell_M + k_s + n_c) = 2048$
MLWE hiding estimate	–	194.408 bits
MSIS binding estimate	–	427.780 bits
Trapdoor quality	α	to be tuned for the sampler
Signature bound	$\beta_{\mathtt{sig}}$	derived from sampler tail bound

Table 1: Primary lattice and commitment parameters.

For this profile,

$$C_M \in R_q, \qquad A_s \in R_q^{1\times 2}, \qquad c = C_M M + A_{s,1}s_1 + A_{s,2}s_2 + e.$$

The binding length is

$$L_{\mathsf{bind}} = 4,$$

and if $\beta_{\mathsf{msg}} = 8$ for all opened small coordinates, the double-opening bound is

$$\beta_{\mathsf{bind}} = 16.$$

D.4 Compact candidate profile

A smaller-ring candidate is shown in Table 2. It keeps the same modulus and coefficient bound, increases k_s, and reduces N.

Knob / Result	Symbol	Value
Ring degree	N	256
Modulus	q	1054721
Semantic message length	ℓ_M	1
Commitment secret length	k_s	4
Commitment target length	n_c	1
Message/randomness bound	B	8
MLWE sample dimension	m_{LWE}	$Nn_c = 256$
MSIS input dimension	m_{SIS}	$N(\ell_M + k_s + n_c) = 1536$
MLWE hiding estimate	–	$\approx$ 194.4 bits

Table 2: Compact candidate profile. A binding estimate must be inserted before using this profile as a full concrete commitment-security profile.

D.5 SmallWood soundness terms

SmallWood's soundness is a combination of DECS polynomial binding, LVCS/PCS binding, PIOP sampling errors, and Fiat–Shamir/grinding terms [7]. For parameters $(N_{\text{tree}}, s, \ell, \ell', \rho, \eta)$, field $\mathbb{F}$, and degree bounds $d_{\text{decs}} = s + \ell - 1$ and d_Q, schematic source-compatible terms include

$$\varepsilon_{\text{decs}} = \binom{N_{\text{tree}}}{d_{\text{decs}} + 2}|\mathbb{F}|^{-\eta} + \frac{Q_{\text{RO}}^2}{2^{2\lambda}}, \qquad \varepsilon_{\text{sum}} = |\mathbb{F}|^{-\rho}, \qquad \varepsilon_{\text{tail}} = \frac{\binom{d_Q}{\ell'}}{\binom{|S|}{\ell'}}.$$

The final knowledge-soundness bound for ARC-SPRUCE must instantiate the exact SmallWood-ARK theorem with the actual row inventory, the actual d_Q, the Fiat–Shamir query bounds, and the grinding parameters. The showing statement includes the bilinear inverse check together with a linear signature equation and the PRF effective degree (Appendix E).

D.6 SmallWood knobs

For the issuance and showing proofs discussed in Section 6, one example parameter tuple is:

$$(s, \ell, \ell', \rho, \theta, \eta) = (16, 25, 2, 2, 4, 19),$$

with masking degree d_Q derived from the instantiated constraint degrees. This tuple is not a final proof-size claim for the ARC-SPRUCE relation.

D.7 Extended benchmark grid

For additional context on how SmallWood knobs affect proof size and proving time, Table 3 reports an example sweep over packing and soundness parameters for standalone SmallWood-style statements. These figures are not a final ARC-SPRUCE proof-size claim unless the row inventory, constraint degrees, PRF checkpoint schedule, and Fiat–Shamir soundness terms are instantiated for the exact ARC relations.

ProofKB	Time (s)	Bits	NCols	ℓ	ℓ'	ρ	θ	η
9.32	0.266	133.44	6	22	2	2	4	17
10.07	0.228	146.93	4	24	2	2	4	17
11.52	0.250	133.22	4	28	2	2	4	17
14.06	0.630	198.01	4	34	8	5	2	26
14.17	0.654	192.26	6	34	8	5	2	26
14.24	0.670	198.48	4	34	8	6	2	26

Table 3: Example sweep over SmallWood knobs for standalone parameter exploration. Sizes exclude public parameters and are not a final ARC-SPRUCE proof-size claim.

E PRF DETAILS AND MISCELLANEOUS

This appendix records a concrete ZK-friendly PRF candidate compatible with the canonical ARC protocol. The construction is Poseidon2-style, but its use as a keyed PRF is an explicit assumption (Assumption 2.27).

E.1 PRF specification

We use a Poseidon2-style permutation with feed-forward and truncation. Let $k \in \mathcal{K}_{\text{key}} \subseteq \mathbb{F}_q^{\ell_{\text{key}}}$ be the secret key encoded by the signed

$\mathbf{k}$, and let $n \in \mathbb{F}_q^{\ell_{\text{nonce}}}$ encode the public nonce. Let $t = \ell_{\text{key}} + \ell_{\text{nonce}}$ be the permutation width. Define

$$\mathsf{F}(k, n) \;=\; \mathrm{Tr}\big(P(k\|n) + (k\|n)\big),$$

where $\mathrm{Tr} : \mathbb{F}_q^t \to \mathbb{F}_q^{\ell_{\text{tag}}}$ outputs the first ℓ_{tag} lanes and P is a Poseidon2-style permutation [8].

E.2 Permutation structure

Let $d \geq 3$ be the smallest constant such that $\gcd(d, q-1) = 1$. Fix round counts R_F (external rounds, even) and R_P (internal rounds), round constants, and MDS matrices M_E, M_I. The permutation has the form

$$P \;=\; E_{R_F} \circ \cdots \circ E_{R_F/2+1} \circ I_{R_P} \circ \cdots \circ I_1 \circ E_{R_F/2} \circ \cdots \circ E_1.$$

Each external round applies the S-box to all lanes, while each internal round applies the S-box to lane 1 only. This structure reduces nonlinear constraints in the post-sign NIZK while retaining diffusion.

External and internal rounds. Let $x = (x_1, \ldots, x_t) \in \mathbb{F}_q^t$. For each external round $i \in [1, R_F]$, fix per-lane constants $(c_1^{(i)}, \ldots, c_t^{(i)}) \in \mathbb{F}_q^t$. For each internal round $i \in [1, R_P]$, fix a scalar constant $\hat{c}^{(i)} \in \mathbb{F}_q$. Then:

$$E_i(x) \;=\; M_E \cdot \big((x_1 + c_1^{(i)})^d, \ldots, (x_t + c_t^{(i)})^d\big)^\top,$$

$$I_i(x) \;=\; M_I \cdot \big((x_1 + \hat{c}^{(i)})^d,\ x_2, \ldots, x_t\big)^\top.$$

The matrices $M_E, M_I \in \mathbb{F}_q^{t\times t}$ are chosen so that they are invertible and provide sufficient diffusion (e.g., Cauchy-style MDS matrices as in Poseidon2).

E.3 Parameter generation

All PRF parameters are public. A typical parameter generator fixes:

- the base field modulus q (common to the rest of the protocol),
- the S-box exponent d with $\gcd(d, q-1) = 1$,
- width $t = \ell_{\text{key}} + \ell_{\text{nonce}}$ and truncation ℓ_{tag},
- round counts R_F, R_P,
- MDS matrices M_E, M_I and round constants $\{c^{(i)}\}, \{\hat{c}^{(i)}\}$.

The truncation length is chosen to satisfy $\ell_{\text{tag}} \geq \lceil \lambda / \log_2(q) \rceil$ as an output-length requirement. Pseudorandomness is not implied by this length condition; it is exactly the concrete assumption stated in Assumption 2.27 for $k \xleftarrow{\$} \mathcal{K}_{\text{key}}$.

E.4 Arithmetization strategy (System A with checkpoints)

Status of the checkpoint schedule. Section 5.5 uses only the abstract interface of three PRF companion families: key binding, nonlinear checkpoint constraints, and final feed-forward/truncation constraints. The grouped schedule recorded in this appendix is a candidate realization for the present instantiation. Any final implementation must fix the exact round counts, checkpoint positions, and resulting effective degree used in the SmallWood soundness calculation.

To keep showing proofs compact, we commit only selected nonlinear checkpoints:

- **Key lanes.** We commit one row per key lane $\mathsf{Key}[i]$ and enforce equality to an encoding of signed $\mathbf{k}$ using selector constraints at fixed Ω-coordinates.
- **S-box outputs.** We commit only the S-box outputs at a checkpoint schedule (e.g., every internal round and periodically among external rounds). Linear wiring between checkpoints (add-round-constants, MDS multiplications) is recomputed by the verifier from opened evaluations and public constants.

This yields a System-A style constraint system where the dominant degree comes from composing at most a small number of unchecked S-box layers between checkpoints.

REFERENCES

[1] C. Baum, I. Damgård, V. Lyubashevsky, S. Oechsner, and C. Peikert. More efficient commitments from structured lattice assumptions. In *Security and Cryptography for Networks*, pages 368–385. Springer, 2018.

[2] S. Brands. Restrictive blinding of secret-key certificates. In *Advances in Cryptology – EUROCRYPT 1995*, pages 231–247, 1995.

[3] J. Camenisch and A. Lysyanskaya. An efficient system for non-transferable anonymous credentials with optional anonymity revocation. In *Advances in Cryptology – EUROCRYPT 2001*, pages 93–118, 2001.

[4] D. Chaum. Security without identification: Transaction systems to make big brother obsolete. *Communications of the ACM*, 28(10):1030–1044, 1985.

[5] A. Dubois, M. Klooß, R. W. F. Lai, and I. K. Y. Woo. Lattice-based proof-friendly signatures from vanishing short integer solutions. Cryptology ePrint Archive, Paper 2025/356, 2025.

[6] T. Espitau, T. T. Q. Nguyen, C. Sun, M. Tibouchi, and A. Wallet. Antrag: Annular ntru trapdoor generation: Making mitaka as secure as falcon. In *Advances in Cryptology – ASIACRYPT 2023*, 2023. Also available as Cryptology ePrint Archive, Paper 2023/1335.

[7] T. Feneuil and M. Rivain. Smallwood: Hash-based polynomial commitments and zero-knowledge arguments for relatively small instances. Cryptology ePrint Archive, Paper 2025/1085, 2025.

[8] L. Grassi, D. Khovratovich, and M. Schofnegger. Poseidon2: A faster version of the poseidon hash function. Cryptology ePrint Archive, Paper 2023/323, 2023.

[9] V. Lyubashevsky, N. K. Nguyen, and M. Plançon. Lattice-based zero-knowledge proofs and applications: Shorter, simpler, and more general. In *Advances in Cryptology – CRYPTO 2022*, pages 71–101. Springer, 2022. Also available as Cryptology ePrint Archive, Paper 2022/284.

[10] S. Schlesinger and J. Katz. Anonymous Credit Tokens. Internet-Draft draft-schlesinger-cfrg-act-00, Internet Engineering Task Force, Aug. 2025. Work in Progress.

[11] C. Yun, C. A. Wood, and A. Faz-Hernandez. Privacy Pass Issuance Protocol for Anonymous Rate-Limited Credentials. Internet-Draft draft-ietf-privacypass-arc-protocol-01, Internet Engineering Task Force, Mar. 2026. Work in Progress.